Tensor Omics (TOX)

Analyze gene expression data and evolutionary relationships between genes

User's Guide

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Part I.

Analysis Recipes

1. Subfunctionalization, Neofunctionalization and Dosage-Effect Detection (SND)

Purpose

Detect genes whose expression trajectories diverge between duplicated or orthologous copies in a way consistent with subfunctionalization, neofunctionalization, or dosage effects, using the Tensor Omics geometric framework.

Maintainer: The researcher currently developing and running SND analysis

Required Input Data

- Per-sample expression matrices (e.g. TPM from `kallisto`), one matrix per species, genes × conditions.
- A sample sheet mapping each column to a condition and replicate.
- Gene-family assignments and ortholog/paralog relationships (produced by the prerequisites below).

Prerequisites / Depends on

- *Gene Family Detection* (see chapter 2)

Note

SND compares copies *within* a gene family, so the family grouping and the ortholog/paralog classification must be finished and sanity-checked before you start here.

Step-by-Step Workflow

Step 1. Normalize each species' expression matrix and assemble a TOX dataset (see the normalization and dataset-creation sub-recipes).

Step 2. Restrict the dataset to families that contain at least two copies in the compared lineages.

Step 3. Compute, per family, the geometric divergence of each copy's expression trajectory relative to the family centroid.

Step 4. Classify each divergent copy as subfunctionalized, neofunctionalized, or dosage-affected using the decision thresholds below.

Step 5. Export the per-gene SND table and the diagnostic trajectory plots.

Example script

Runs the full SND analysis on the bundled example data and writes an `snd_results.tsv` table plus per-family trajectory plots. Running it unchanged reproduces the default results referenced in this recipe.

`python/examples/snd_detection.py`

```
# --- input data
    ------------------------------------------------------------
species = {
    "ath": "data/ath_tpm.tsv",    # Arabidopsis thaliana
    "aly": "data/aly_tpm.tsv",    # Arabidopsis lyrata
}
sample_sheet = "data/samples.tsv"

# --- parameters (see the parameter table below)
    -----------------------------
min_family_size   = 2
divergence_cutoff = 0.35      # angular divergence from the family
    centroid
dosage_log2fc     = 1.0       # |log2 fold-change| flagged as a dosage
    effect
```

Listing 1.1: Fragments to edit: input paths and thresholds

Important Parameters

Parameter	Controls	When / which values
`min_family_size`	Smallest number of copies a family must have to be tested	Keep at 2 for pairwise comparisons; raise to focus on larger, higher-confidence families
`divergence_cutoff`	Angular divergence from the family centroid above which a copy is called "divergent"	Lower (~0.2) for sensitive screening; raise (~0.5) to report only strong divergence
`dosage_log2fc`	Absolute log2 fold-change flagged as a dosage effect	Use `1.0` (2×) as a default; tighten for deep, low-noise data

Expected Results

After step 3 you should have one divergence score per copy, most near zero. On the example data roughly 5–10% of copies exceed `divergence_cutoff`; a much larger fraction usually indicates a normalization or batch problem rather than genuine biology.

Common Pitfall

Comparing raw TPM across species without a common normalization inflates divergence scores. Always assemble the TOX dataset from consistently normalized matrices before computing divergence.

Pitfall: unbalanced replicates

Families whose copies are measured in different conditions produce artefactual trajectory divergence. Restrict to the shared condition set, or impute with care, before classification.

Interpretation

A copy with high divergence but a conserved expression *shape* relative to a sibling suggests *dosage* change; divergent shape with retained breadth suggests *subfunctionalization*; a novel expression domain absent from all other copies suggests *neofunctionalization*. Read the per-family trajectory plots alongside the table before drawing conclusions.

References

- Robinson, McCarthy & Smyth (2010). edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics* 26(1):139–140.
- Emms & Kelly (2019). OrthoFinder: phylogenetic orthology inference for comparative genomics. *Genome Biology* 20:238.

Part II.

Shared Sub-Recipes

2. Gene Family Detection

Purpose

Group protein-coding genes across species into gene families so that downstream analyses can compare orthologous and paralogous copies. This is a shared preprocessing step reused by several analysis recipes.

Maintainer: The researcher who last ran family detection for the dataset

Required Input Data

- One protein FASTA per species (primary transcripts only).
- A species list / directory layout expected by OrthoFinder.

Step-by-Step Workflow

Step 1. Run OrthoFinder on the per-species protein FASTAs to obtain orthogroups.

Step 2. Optionally refine large orthogroups with MCL clustering to split fused families.

Step 3. Emit a gene-to-family table used by downstream recipes.

Example script

Wraps OrthoFinder + optional MCL refinement and writes `orthogroups.tsv`. A default run on the example proteomes reproduces the family table used elsewhere in this manual.

`scripts/gene_family_detection.sh`

Important Parameters

Parameter	Controls	When / which values
`inflation (-I)`	MCL granularity when refining orthogroups	Higher (`2.0`–`4.0`) yields smaller, tighter families; lower yields larger, more inclusive families

Common Pitfall

Including multiple splice isoforms per gene inflates family sizes and biases every downstream comparison. Reduce each proteome to primary transcripts first.

References

- Emms & Kelly (2019). OrthoFinder: phylogenetic orthology inference for comparative genomics. *Genome Biology* 20:238.